

Python File Input/Output

1. What is File I/O?

- File I/O is the process of reading data from a file and writing data to a file using a program.
- I → Input → Reading from a file
- O → Output → Writing to a file

2. Types of Files

1) Text Files

- .txt
- .csv
- .json

2) Binary Files

- .jpg
- .png
- .mp3

3. open() Function

- open() is used to open a file and create a connection between Python and the file.

```
file = open("filename", "mode")
```

```
file = open("student.txt", "r")
```

4. File Modes

- Mainly use

Mode	Meaning
r	Read
w	Write
a	Append
x	Create

5. r — Read Mode

- Used to read existing data.

```
file = open("student.txt", "r")  
  
data = file.read()  
  
print(data)  
  
file.close()
```

6. w — Write Mode

- Used to write data to a file.

```
file = open("student.txt", "w")  
  
file.write("Name: Rahul")  
file.close()
```

7. a — Append Mode

- Used to add new data **at the end** of the file.

```
file = open("student.txt", "a")  
  
file.write("\nAge: 22")  
  
file.close()
```

8. x — Create Mode

- Used to create a **new file**.

```
file = open("student.txt", "x")  
file.close()
```

9. write()

- Used to write a string into a file.

```
file = open("data.txt", "w")  
  
file.write("Hello Python")  
  
file.close()
```

Method	Returns / Purpose
write()	Writes a single string to the file
writelines()	Writes multiple strings to the file

10. read()

- Reads the entire file.

```
file = open("student.txt", "r")

data = file.read()

print(data)

file.close()
```

Method	Returns
read()	Entire content as string
readline()	One line as string
readlines()	All lines as list

11. close()

- After working with a file, close it.

```
file.close()
```

12. with open()

- with open() automatically manages the file and closes it after use.

```
with open("student.txt", "r") as file:

    data = file.read()

    print(data)
```